Ethan O'Farrell

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Work Experience

Bioinformatician – Michigan Neuroscience Institute **January 2023 - present**

- Processing and analyzing **multiomic single cell RNA and ATAC-seq** datasets of 500,000+ cells with cellranger and the scanpy ecosystem, combining novel techniques in omics integration, celltype-specific analysis, and differential expression testing. Work resulted in 3 publications in progress on gene regulation and psychedelic therapeutics
- Identifying putative genomic enhancer regions with bulk epigenomic sequencing, ChromHMM analysis, edgeR, and bedtools to **prioritize enhancer candidates for wet-lab experiments**
- Building and deploying **web apps** for visualizing high-dimensional RNA sequencing data with Flask, D3.js, and IGV.js, integrating SQLite databases and HDF5 storage. Apps automate and reduce figure generation time by 10x
- Developed an **efficient Maximum Likelihood Estimation method** for the odds ratio parameter of computationally expensive non-central hypergeometric distributions. Methods reduced runtime from exponential to linear, enabling usage for **statistical tests for set overlap** in genomic analysis
- Standardizing **genomic data processing pipelines** and analysis by building modular rules with Snakemake, reducing pipeline setup time from 5 hours to 5 minutes
- Author of the **R package kwanlibR**, introducing methods for automating and standardizing differential ATAC-seq analysis and improving visualizations for differential expression

Data Scientist – Milliman **September - December 2022**

- Preprocessed client data with PySpark on Databricks to train a XGBoost **regression algorithm** for sales leads
- Maintained **dashboards** in R Shiny by adding interactive Plotly visualization, anonymous data download capabilities, debouncing user inputs, and modernizing interfaces, serving insights to 11 industry partners

Junior Data Scientist – Jane Marketplace **May - August 2021**

- Developed a **complementary product recommendation pipeline** using TensorFlow and a modified Siamese neural network to increase user cart sizes by 10% on average
- Conducted an **explainability analysis of daily revenue** with generalized linear models and Seaborn visualizations to discover potential boosts of $50,000 daily revenue from data-driven product selection

Business Intelligence Developer – Nuclear Promise X **September - December 2020**

- Built Power BI **dashboards monitoring 65 performance metrics** of a nuclear power station for senior employees to track KPI progress
- Led development for a beach rip current safety initiative POC with a Python-based Raspberry Pi and JavaScript website compiling and exposing weather data through **REST APIs**

Education

University of Toronto – Master of Science in Applied Computing, Data Science for Biology **2025 - 2026**

- Relevant courses – Readings in Bioinformatics, Artificial Intelligence for Drug Discovery, Theory and Methods for Complex Spatial Data, Natural Language Computing, Algorithms for Private Data Analysis

University of Waterloo – Bachelor of Mathematics, Data Science **2018 - 2023**

- 2024 Co-op Student of the Year, Honourable Mention
- 2018 to 2022 Men's Golf Team, 2019 Men's Rowing Captain

Technical Skills

Languages	Python, R, SQL, JavaScript, HTML/CSS, Bash
Computation	numpy, pandas, scikit-learn, scipy, tensorflow, tidyverse, sqlite, scanpy
Visualization	matplotlib, seaborn, plotnine, plotly, ggplot, leaflet, D3.js, Power BI
Tools	git, flask, shiny, spark, slurm, snakemake, airflow, AWS Sagemaker, Azure Databricks, MLops principles